

8	(a) energy to separate nucleons (in a nucleus) separate to infinity	M1 A1	[2]
	(b) (i) fission	B1	[1]
	(ii) 1. U: near right-hand end of line	B1	[1]
	2. Mo: to right of peak, less than $\frac{1}{3}$ distance from peak to U	B1	[1]
	3. La: $0.4 \rightarrow 0.6$ of distance from peak to U	B1	[1]

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- (iii) 1. right-hand side, mass = 235.922 u
mass change = 0.210 u
C1
A1 [2]
2. energy = mc^2
 $= 0.210 \times 1.66 \times 10^{-27} \times (3.0 \times 10^8)^2$
 $= 3.1374 \times 10^{-11} \text{ J}$
 $= 196 \text{ MeV}$ (need 3 s.f.)
C1
A1 [3]
- (use of 1 u = 934 MeV, allow 3/3; use of 1 u = 930 MeV or 932 MeV, allow 2/3)*
(use of 1.67×10^{-27} not 1.66×10^{-27} scores max. 2/3)

Section B